

✅ FULL ANALYSIS: How Warp Handles Swarms / Multi-Agent Orchestration

Repository: <https://github.com/warpgdotdev/warp> (open-sourced April 28, 2026)

Current Version: Latest master (as of 2026-04-30)

Key Finding: Warp does not have a native, local “swarm” system like swarms-rs, AutoGen, or CrewAI. Instead, it uses a **centralized, cloud-first orchestration layer called Oz** to coordinate multiple agents.

1. Official Terminology & Architecture (Direct from Repo)

Oz = Warp’s Multi-Agent Orchestration Platform

- Oz is explicitly described as the “**cloud agent orchestration platform**”.
- The README and dashboard (build.warp.dev) repeatedly refer to “**thousands of Oz agents**” working together on real tasks.
- Core Oz workflow (the “swarm-like” behavior):
 - **Triage issues**
 - **Write specs**
 - **Implement changes (code)**
 - **Review PRs**
 - **Merge / ship**

This is the closest thing Warp has to a “swarm”: a structured, multi-stage, multi-agent pipeline managed by Oz.

How the Coordination Actually Works

- **Skills System** (`.agents/skills/` directory + `resources/bundled/skills/`)
Agents are modular. Each skill is a self-contained prompt + tool set. Agents can chain skills.
- **Command Blocks**
Persistent, AI-assisted blocks inside the terminal that can represent multi-step workflows.

- **Oz Dashboard** (build.warp.dev)

Central monitoring plane where you watch hundreds/thousands of Oz agents in parallel. It shows in-flight work, top contributors, and lets you click into live agent sessions.

- **Contribution Workflow**

Community + Oz agents collaborate on the Warp repo itself using this exact flow (triage → spec → code → review). The open-source client is designed to participate in Oz-driven development.

Important Distinction

- The **client** (the terminal you use) is open-source and local.
- The **heavy orchestration, state management, and multi-agent coordination** happen in **Oz**, which is still a cloud service (app.warp.dev backend).
- You can bring your own CLI agents (Claude Code, Codex, Gemini CLI, local models), but the swarm-style coordination is routed through Oz.

2. Relevance to Sovereign Machina (Full Cluster Context)

Strengths We Can Mine

- The **Oz-style structured workflow** (triage → spec → implement → review) is exactly what we already designed for Hermes' "Oz contribution workflow".
- The **skills directory** (`.agents/skills/`) is a perfect match for our Profile Framework's `sidecars/skills/` extension.
- Command blocks and persistent workflows are excellent patterns for the Warp observer bridge.

Limitations for Our Use Case

- Oz is **cloud-centric** (state, routing, multi-agent memory live in Warp's backend).
- No local swarm execution engine in the open-source client.
- We cannot (and should not) use the AGPL parts of Oz directly.

How We Incorporate This into Sovereign Machina

We treat Oz as a **reference implementation** of the exact live-coding-partner workflow you want:

- Our `sovereign-warp-observer` sidecar + Hermes Supervisor will replicate the Oz flow **locally and sovereignly**.
- Warp (your dev workspace) sends telemetry → Observer → MemPalace → GEPA/HALO → suggestions back to Warp + Pretext HUD.
- This gives you the same “thousands of agents” feel, but fully local, fully under your control, and running on the full quaternary mesh.

Live Coding Partner Acceleration

With this mined Oz pattern, Hermes can now:

- Watch your Warp session in real time.
- Automatically triage what you are working on.
- Generate specs or refactors.
- Suggest next actions as rich blocks in Warp.
- Evolve its own skills based on your Gemini/Droid + Warp activity.

This is one of the strongest ways to make Hermes a true live partner without merging runtimes.

NEXT STEPS (Your Call)

Reply with the number you want first:

1. **Updated** `sovereign-warp-observer` **spec** incorporating the Oz workflow patterns (with starter code).
2. **Full Oz-style contribution workflow** for Hermes (triage → spec → implement → review) as a new sidecar.
3. **Profile.yaml** extension that adds Oz-style `orchestration_policy` for dev sessions.
4. **All three** as one complete package.

This Warp + Oz mining round is extremely high-signal and directly accelerates the live coding partner vision you described.

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What would you like to initiate?